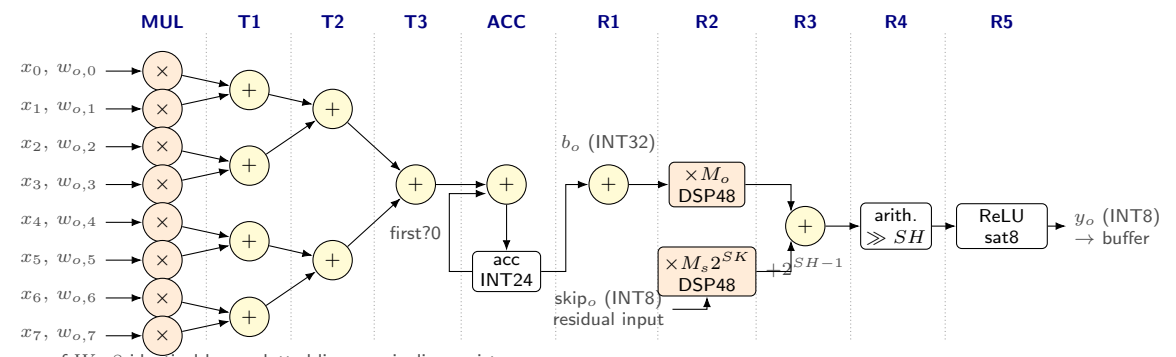

MUL
T1
T2
T3
ACC
R1
R2
R3
R4
R5
$x_0$, $w_{o,0}$
$x_1$, $w_{o,1}$
$x_2$, $w_{o,2}$
$x_3$, $w_{o,3}$
$x_4$, $w_{o,4}$
$x_5$, $w_{o,5}$
$x_6$, $w_{o,6}$
$x_7$, $w_{o,7}$
first?0
acc
INT24
$b_o$ (INT32)
$\times M_o$
DSP48
$\times M_s 2^{SK}$
DSP48
skip$_o$ (INT8)
residual input
$+2^{SH-1}$
arith.
$\gg SH$
ReLU
sat8
$y_o$ (INT8)
$\rightarrow$ buffer
one of $W$=8 identical lanes; dotted lines = pipeline registers